

AI-Powered Driver Wellness & Safety Monitoring System

Five deep-learning modules — each benchmarked on a strictly held-out split, with error diagnosis, efficiency measurement and deployment-readiness assessment.



Kushagra



Shiwani



Shubham



Sohini



Ravina

From Trained Models to Held-Out Evaluation

Milestone 4 delivered five trained checkpoints. Milestone 5 measures each on data it never saw — impartially, then diagnostically.



Benchmark impartially

Score every frozen checkpoint on a strictly held-out, subject- or driver-disjoint test split.



Diagnose failures

Go beyond a single accuracy number — build confusion matrices and per-class breakdowns.



Root-cause each error

Tie every failure mode to an interpretable cause: label noise, small objects, class collapse.



Measure efficiency

Record parameters, FLOPs and per-frame latency on the shared Tesla T4 evaluation GPU.



Assess readiness

Judge each module against the accuracy / speed / memory trade-off for edge deployment.



Map improvements

Convert every diagnosed gap into a concrete, prioritised short- and long-term action.

Single Frames Cannot Explain Fatigue

Fatigue and distraction are leading causes of commercial-fleet accidents — yet fleet platforms have **no continuous, objective view of driver state**. One cabin camera can observe it all.

WHY THE OLD APPROACH FAILS

- A blink and drowsiness look identical in one frame
- Static frame counts and hardcoded thresholds
- Talking, singing or laughing trigger false alerts
- Frame-level YOLO has no sense of time

THE REVISED SYSTEM · MILESTONES 1–5

- Learns fatigue as it develops across video sequences
- Temporal deep models: CNN-LSTM / BiLSTM, LSTM, TCN
- MediaPipe / OpenCV give auxiliary features, not decisions
- YOLO reserved for object-level safety compliance
- Fusion layer → one Wellness Score: Safe / Caution / High Risk

Five Datasets, One Wellness System

Each module was cleaned, validated and split leakage-free (subject- / sequence-wise) in Milestone 2.

MODULE	DATASET	SCALE & CLEANING
 Video-Based Fatigue	UTA-RLDD	60 subjects · 182 drowsiness videos · 3 fatigue classes
 Landmark Fatigue	YawDD	349 videos → 286,289 valid frames · 99.35% extraction
 Driver Activity	AUC Distracted Driver	4,307 clean images · 5 activity classes
 Seat Belt & Phone	DMS Dataset	6,140 images · 6,701 boxes · 2 classes
 Smoking & Drinking	Merged YOLO set	12,909 images · 15,900 instances · 2 classes

Model Architectures & Experiments

Shared method: shortlist candidates → filter by edge latency & memory → baseline on the module’s data → lock full spec.

MODULE	CANDIDATES	SELECTED	KEY EXPERIMENT
Video Fatigue	CNN-LSTM · GRU · TCN · 3D-CNN	EfficientNet-B0 + BiLSTM	16-frame clips; spatial + temporal
Landmark Fatigue	LSTM · GRU · TCN · MLP	LSTM	swept lr / bs / hidden / window (45×5)
Driver Activity	MobileNetV3 · ResNet50 · EfficientNet-B0	MobileNetV3	EffNet 94% · chose edge · tuned lr 5e-4
Seat Belt & Phone	YOLOv8n · YOLO11n · YOLOv8s	YOLOv8n	nano fastest on Jetson / Pi @ 640 ²
Smoking & Drinking	YOLOv8n · YOLO11n · YOLOv8s	YOLOv8n	6-trial guided grid → lr .001 AdamW cos

Final Trained Models

Five trained checkpoints — the inputs Milestone 5 now benchmarks on strictly held-out data.

K

Video Fatigue

EfficientNet-B0 + BiLSTM

Baseline val acc
31.8%

longer training flagged

S

Landmark Fatigue

LSTM

3-class risk mapping

Normal / Talking / Yawning

S

Driver Activity

MobileNetV3

~94% val accuracy

12.5 ms/frame · edge-ready

S

Seat Belt & Phone

YOLOv8n

mAP@50 95.26%

P 92.9 · R 91.2

R

Smoking & Drinking

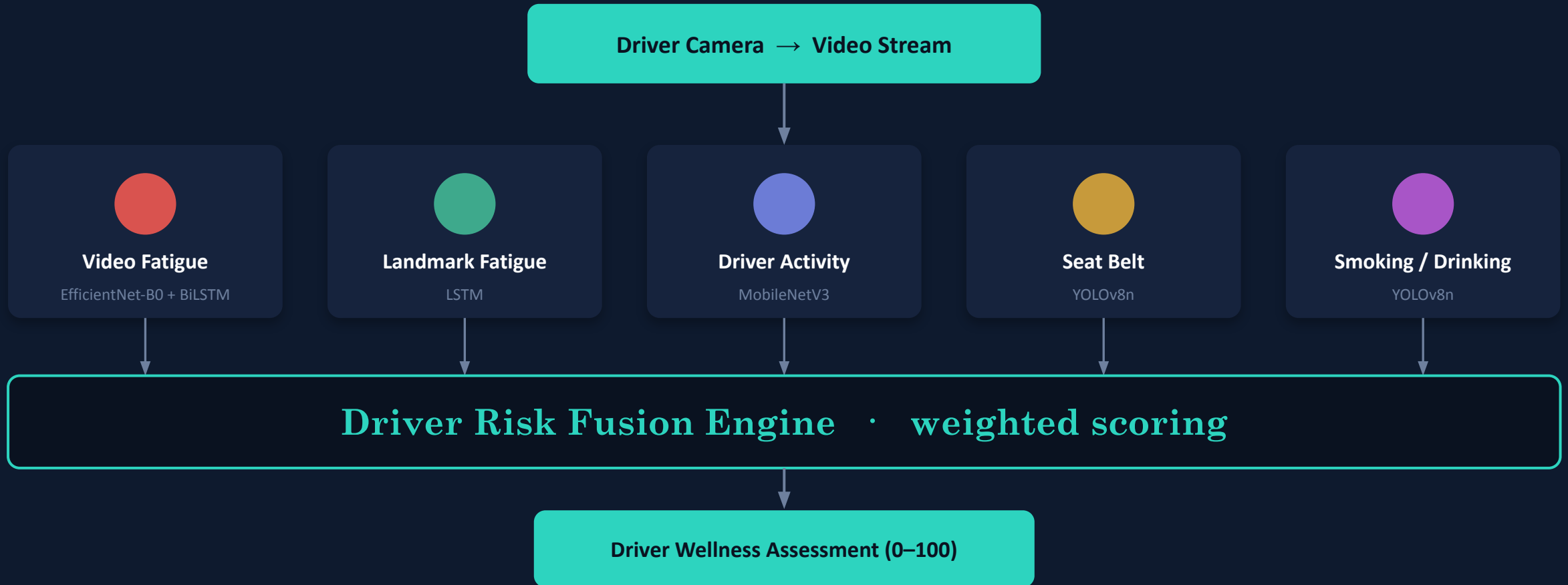
YOLOv8n

mAP@50 0.818

mAP@50-95 0.451 · 3.01M

Driver Risk Fusion Engine → weighted scoring → Driver Wellness Score (0–100)

Five Models, One Wellness Score



Five Modules, Five Held-Out Evaluations

OWNER	MODULE	MODEL	TEST SET	HEADLINE RESULT
 Kushagra	Video-Based Fatigue	EfficientNet-B0 + BiLSTM	7,117 seq · 6 subj	Acc 33.6%
 Shiwani	Landmark Fatigue	LSTM (EAR·MAR·pose)	1,400 windows	Acc 64.1%
 Shubham	Driver Activity	MobileNetV3-Large	1,093 images	Acc 93.1%
 Sohini	Seat Belt & Phone	YOLOv8n	cabin image set	mAP@50 0.95
 Ravina	Smoking & Drinking	YOLOv8n	371 img · 445 box	mAP@50 0.82

Each module is evaluated inference-only on a frozen checkpoint against a strictly held-out split.

*Subject-disjoint indexed test set***MODEL & CONFIGURATION**

Backbone **EfficientNet-B0**

Temporal layer **Bidirectional LSTM**

Hidden size **256**

Sequence length **16 frames**

Classes **Safe · Caution · High-Risk**

Parameters **7.16 M**

Checkpoint **driver_wellness_epoch2.pth**

HELD-OUT TEST SET**7,117**

indexed sequences

6

unseen subjects

≈balanced

3-class support

WHAT THIS EVALUATION TESTS

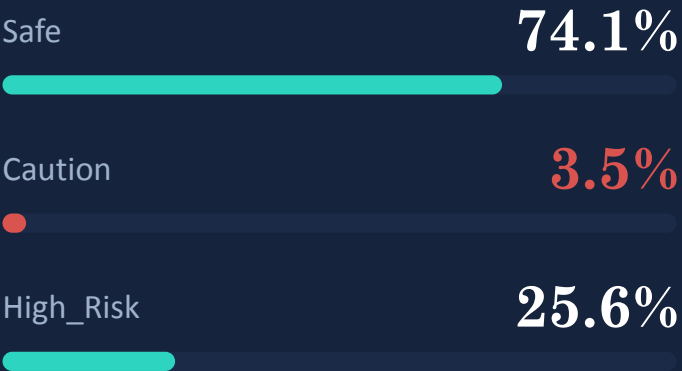
Whether a frozen 2-epoch checkpoint generalises to unseen drivers, and — critically — whether the three fatigue classes stay separable or collapse. The fixed-seed subject-disjoint split the indexed split preserved subject-disjoint separation across train/validation/test.

K

Kushagra

Held-out test evaluation

PER-CLASS RECALL



Baseline: random 33.3% · majority 36.2%

QUANTITATIVE RESULTS — HELD-OUT TEST



READING & NEXT STEP

The model barely exceeds uniform-random and stays below the majority-class baseline. The dominant failure is severe class collapse — Caution recall of 3.5% and two-thirds of High-Risk sequences predicted as Safe. It is therefore an experimental signal for the fusion engine, not a standalone safety trigger; longer temporal context and class-sensitive training are the next levers.

*Interpretable landmark features***MODEL & CONFIGURATION**

Model	2-layer LSTM
Features	EAR · MAR · Pitch·Yaw·Roll
Window size	45 frames
Extractor	MediaPipe Face LandMarker
Classes	Normal · Talking · Yawning
Loss	Weighted CE
Checkpoint	m4_lstm_full_final.pt

HELD-OUT TEST SET**1,400**

test windows

11.3%

Yawning (minority)

subject

disjoint split

WHAT THIS EVALUATION TESTS

Whether hand-engineered facial signals separate the three behaviours, and whether the fatigue-state decision threshold can be empirically justified rather than left as an unvalidated heuristic. Inputs are human-interpretable, so any prediction is cross-checkable against the raw MAR trace.



Held-out test + threshold tuning

LABEL CORRECTION IMPACT

	ACCURACY	MACRO-F1
Before fix	40.8%	38.4%
After fix	64.1%	61.8%

+23.3 pts accuracy from the 3-class label restructuring

CORRECTED 3-CLASS TEST RESULTS

64.1%	0.62	0.61	94.4%
Accuracy	Macro-F1	Yawning AP	Sensitivity

READING & THRESHOLD DECISION

The dominant error (Talking→Normal, 53% of mistakes) is video-level label granularity, not model capacity. Cross-class Talking↔Yawning confusion is minor. Because a missed drowsy driver is the costly failure and this module is only one of five fused inputs, the Drowsy threshold is revised from 15% to 5% — lifting sensitivity to 94.4% at an acceptable false-alarm cost.



Shubham

MODEL & CONFIGURATION

Architecture	MobileNetV3-Large
Training	From scratch
Input	224×224 RGB
Optimizer	AdamW · lr 3e-4
Classes	5 activities
Parameters	4.21 M
Best val acc	94.4%

HELD-OUT TEST SET

1,093

test images

5

activity classes

driver

disjoint split

WHAT THIS EVALUATION TESTS

Whether a lightweight, edge-oriented classifier can recognise five driving behaviours — safe driving, texting, talking on phone, turning, and a catch-all other — accurately enough for real-time deployment, and where the intra-class variance of the catch-all class costs accuracy.

s

Shubham

Held-out test + model comparison

PER-CLASS F1-SCORE



OVERALL HELD-OUT PERFORMANCE



WHY MOBILENETV3 — ACCURACY VS EFFICIENCY

MODEL	ACC	PARAMS	SPEED
MobileNetV3 (ours)	93.1%	4.2 M	12.5 ms
EfficientNet-B0	94.0%	5.3 M	28.7 ms
ResNet50	92.1%	25.5 M	45.2 ms



YOLOv8n + inference safeguards

MODEL & CONFIGURATION

Detector	YOLOv8n
Classes	Seatbelt · Phone
Input	1280×1280 letterbox
Gamma correction	$\gamma = 1.4$
Temporal filter	≈85% frame stability
NMS	Class-agnostic
Speed	6.8–10.4 ms/frame

EVALUATION DATASET

day+night	glare	2
cabin images	& heavy shadow	annotated classes

WHAT THIS EVALUATION TESTS

Whether a small detector plus a stateful temporal-consensus pipeline can hold up on continuous cabin video — beyond static-image accuracy — resisting flicker, arm/shadow confusion (the "seesaw" effect), and out-of-distribution lighting.



Validation metrics + mitigations

DEPLOYMENT SAFEGUARDS

- **Decoupled thresholds**

Seatbelt 0.40 · Phone 0.10

- **Class-agnostic NMS**

removes overlapping arm/shadow boxes

- **Spatial filtering**

rejects window-glass & tiny boxes

- **Temporal consensus**

~85% stability before confirming

YOLOV8N DETECTION METRICS

0.953

mAP@50

0.673

mAP@50–95

0.937

Precision

0.903

Recall

READING & LIMITATIONS

Static detection is strong (mAP@50 = 0.95), but continuous-stream deployment exposes flicker and lighting sensitivity that single-image scoring hides. Heavy shadow can hide the phone; strong glare washes out the belt. The inference safeguards resolve the arm-vs-belt "seesaw"; the metrics come from the notebook validation set rather than a fully separate held-out split — flagged as the key caveat.

R

Ravina

Strictly held-out test split

MODEL & CONFIGURATION

Detector	YOLOv8n
Classes	Smoking · Drinking
Input	640×640
Training	80 epochs · AdamW
LR schedule	cosine · lr0 1e-3
Parameters	3.01 M
Compute	8.1 GFLOPs

HELD-OUT TEST SPLIT

371

test images

445

labelled boxes

frozen

yolov8n_best.pt

WHAT THIS EVALUATION TESTS

Impartial benchmarking of a frozen checkpoint on a split that contributed nothing to training, plus root-cause diagnosis of every failure and a check that validation and test scores agree — the signature of clean splits and no leakage.

R Ravina

Held-out test + per-class split

THE CLASS IMBALANCE

recall by class

Smoking **0.93**



Drinking **0.67**



≈0.26 recall gap — drinking events are far more likely to be missed.

OVERALL HELD-OUT TEST METRICS

0.85

Precision

0.80

Recall

0.82

mAP@50

0.45

mAP@50–95

READING — A SAFELY-DIRECTED FAILURE PROFILE

Val and test mAP@50–95 match within 0.005 — clean splits, genuine generalisation. Of 46 failures only 6 are true cross-class confusion; the rest are missed or low-confidence boxes. The model fails the safe way — it declines to fire rather than asserting a confident wrong class — which temporal smoothing can recover. Drinking recall and strict localisation are the gaps to close.

Five Evaluations, One Scorecard

MODULE	HEADLINE	STRENGTH	PRIMARY GAP
● Video Fatigue	Acc 33.6%	Streaming pipeline built	Severe class collapse
● Landmark Fatigue	Acc 64.1%	Interpretable + validated threshold	Talking label granularity
● Driver Activity	Acc 93.1%	Fast, edge-ready, high AUC	Catch-all class variance
● Seat Belt & Phone	mAP 0.95	Robust inference safeguards	Lighting / temporal flicker
● Smoking & Drinking	mAP 0.82	Safe failure, no leakage	Drinking recall (0.67)

Across the board, errors are diagnosed and mapped to concrete fixes — no gap is left unexplained.